Polynomial WS 3

1. If α and β are the roots of the quadratic equation $x^2 - 4x + 2 = 0$, find:
   a. $\alpha + \beta$
   b. $\alpha\beta$
   c. $\alpha^2\beta + \alpha\beta^2$
   d. $\frac{1}{\alpha} + \frac{1}{\beta}$
   e. $(\alpha + 2)(\beta + 2)$
   f. $\alpha^2 + \beta^2$
   g. $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$
   h. $\alpha\beta^3 + \alpha^3\beta$
   i. $\left(\alpha + \frac{1}{\alpha}\right)\left(\beta + \frac{1}{\beta}\right)$

2. If α, β and γ are the roots of the equation $x^3 + 2x^2 - 11x - 12 = 0$, find:
   a. $\alpha + \beta + \gamma$
   b. $\alpha\beta + \alpha\gamma + \beta\gamma$
   c. $\alpha\beta\gamma$
   d. $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$
   e. $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$
   f. $(\alpha + 1)(\beta + 1)(\gamma + 1)$
   g. $(\alpha\beta)^2\gamma + (\alpha\gamma)^2\beta + (\beta\gamma)^2\alpha$
   h. $\alpha^2 + \beta^2 + \gamma^2$
   i. $(\alpha\beta)^{-2} + (\alpha\gamma)^{-2} + (\beta\gamma)^{-2}$

   Now find the roots of the equation $x^3 + 2x^2 - 11x - 12 = 0$ by factoring the LHS. Hence check your answers for expressions (a)-(i).

3. If α, β, γ and δ are the roots of the eqation $x^4 - 5x^3 + 2x^2 - 4x - 3 = 0$, find:
   a. $\alpha + \beta + \gamma + \delta$
   b. $\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta$
   c. $\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta$
   d. $\alpha\beta\gamma\delta$
   e. $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$
   f. $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\alpha\delta} + \frac{1}{\beta\gamma} + \frac{1}{\beta\delta} + \frac{1}{\gamma\delta}$
   g. $\frac{1}{\alpha\beta\gamma} + \frac{1}{\alpha\beta\delta} + \frac{1}{\alpha\gamma\delta} + \frac{1}{\beta\gamma\delta}$
   h. $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$

4. If α and β are the roots of the equation $2x^2 + 5x - 4 = 0$, find:
   a. $\alpha + \beta$
   b. $\alpha\beta$
   c. $\alpha^2 + \beta^2$
   d. $\alpha^3 + \beta^3$
   e. $|\alpha - \beta|$

5. Given the polynomial $P(x) = x^3 - x^2 - x + 10$.
   a. Show that -2 is a zero of $P(x)$.
   b. Given that the zeroes of $P(x)$ are $-2, \alpha$ and β, show that $\alpha + \beta = 3$ and $\alpha\beta = 5$.
   c. Solve simultaneously the two equations in part (b) (you will need to form a quadratic in α), and hence show that there are no such real numbers α and β.
   d. Hence state how many times the graph of the cubic crosses the x-axis.

6. a. Suppose that $x - 3$ and $x + 1$ are factors of $x^3 - 6x^2 + ax + b$. Find a and b, and hence use sum and product of zeroes to find the other zero.
   b. Suppose that $2x^3 + ax^2 - 14x + b$ has zeroes at -2 and 4. Find a and b, and hence use sum and product of zeroes to find the other zero.

7. a. Find values of a and b for which $x^3 + ax^2 - 10x + b$ is exactly divisible by $x^2 + x - 12$, and then factor the cubic.
   b. Find values of a and b for which $x^2 - x - 20$ is a factor of $x^4 + ax^3 - 23x^2 + bx + 60$, and then find all the zeroes.

8. The polynomial $P(x) = x^3 - Lx^2 + Lx - M$ has zeroes $\alpha, \frac{1}{\alpha}$ and β.
   a. Show that:
      i. $\alpha + \frac{1}{\alpha} + \beta = L$
      ii. $1 + \alpha\beta + \frac{\beta}{\alpha} = L$
      iii. $\beta = M$
   b. Show that either $M = 1$ or $M = L - 1$.

9. The cubic equation $x^3 - Ax^2 + 3A = 0$, where $A > 0$, has roots α, β and $\alpha + \beta$.
   a. Use the sum of the roots to show that $\alpha + \beta = \frac{1}{2}A$.
   b. Use the sum of the products of pairs of roots to show that $\alpha\beta = -\frac{1}{4}A^2$.
   c. Show that $A = 2\sqrt{6}$

10. a. Find the roots of the equation $4x^3 - 8x^2 - 3x + 9 = 0$, given that two of the roots are equal. [HINT: Let the roots be α, α and β.]
    b. Find the roots of the equation $3x^3 - x^2 - 48x + 16 = 0$, given that the sum of two of the roots is zero. [HINT: Let the roots be $\alpha, -\alpha$ and β]
    c. Find the roots of the equation $2x^3 - 5x^2 - 46x + 24 = 0$, given that the product of two of the roots is 3. [HINT: Let the roots be $\alpha, \frac{3}{\alpha}$ and β.]
    d. Find the zeroes of the polynomial $P(x) = 2x^3 - 13x^2 + 22x - 8$, given that one zero is the product of the other two. [HINT: Let the zeroes be α, β and $\alpha\beta$.]

11. a. Find the roots of the equation $9x^3 - 27x^2 + 11x + 7 = 0$, if the roots form an arithmetic sequence. [HING: Let the roots of be $\alpha - d$, α and $\alpha + d$.] Then find the point of inflexion of $y = 9x^3 - 27x^2 + 11x + 7$, and show that its x —coordinate is one of the roots.
    b. Find the zeroes of the polynomial $P(x) = 8x^3 - 14x^2 + 7x - 1$, if the zeroes form a geometric sequence, [HINT: Let the zeroes be $\frac{\alpha}{r}, \alpha$ and αr.]
    c. Solve the equation $2x^3 - 3x^2 - 3x + 2 = 0$, given that the roots are in geometric progression.

12. a. Two of the roots of the equation $x^3 + 3x^2 - 4x + a = 0$ are opposites. Find the value of a and the three roots.
    b. Two of the roots of the equation $4x^3 + ax^2 - 47x + 12 = 0$ are reciprocals. Find the value of a and the three roots.
    c. Find α and β if the zeros of the polynomial $x^4 - 3x^2 - 8x^2 + 12x + 16 = 0$ are $\alpha, 2\alpha, \beta$ and 2β.

13. a. If one root of the equation $x^3 - bx^2 + cx - d = 0$ is equal to the product of the other two, show that $(c + s)^2 = d(b + 1)^2$.
    b. If the roots of the equation $x^3 + ax^2 + bx + c = 0$ form an arithmetic sequence, show that $9ab = 2a^3 + 27c$, and that one of the roots is $-\frac{1}{3}a$.
    c. If the roots of the equation $x^3 + ax^2 + bx + c = 0$ form a geometric sequence, show that $b^3 = a^3c$, and that one of the roots is $-\sqrt[3]{c}$.
    d. The polynomial $P(x) = x^4 + ax^3 + bx^2 + cx + d$ has two zeroes which are opposites and two zeroes which are reciprocals. Show that:
       i. $b = 1 + d$
       ii. $c = ad$
    e. If the zeroes of the cubic $x^3 + ax^2 + bx + c = 0$ form an arithmetic sequence, show that the point of inflexion lies on the x-axis.

14. Consider the cubic polynomial $P(x) = ax^3 + bx^2 + bx + cx + d$, which has zeroes α, β and γ such that $\gamma = \alpha + \beta$.
    a. Show that $\alpha + \beta = -\frac{b}{2a}$
    b. Show that $\alpha\beta = \frac{2d}{b}$.
    c. Hence show that α and β are also the roots of the equation $2abc^2 + b^2x + 4ad = 0$

15. a. The cubic equation $2x^3 - x^2 + x - 1 = 0$ has roots α, β and γ. Evaluate:
    i. $\alpha + \beta + \gamma$
    ii. $\alpha\beta + \alpha\gamma + \beta\gamma$
    iii. $\alpha\beta\gamma$
    iv. $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$
    b. The equation $2\cos^3\theta - \cos^2\theta + \cos\theta - 1 = 0$ has roots $\cos a$, $\cos b$ and $\cos c$. Using the results in (a), prove that $\sec a + \sec b + \sec c = 1$.

16. a. Show that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$
    b. Show that the cubic equation $8x^3 - 6x + 1 = 0$ reduces to the form $\cos 3\theta = -\frac{1}{2}$ by substituting $x = \cos\theta$.
    c. Hence Find the three solutions to the cubic equation.
    d. Use the sum and product of roots to evaluate:
    i. $\cos\frac{2\pi}{9} + \cos\frac{4\pi}{9} - \cos\frac{\pi}{9}$
    ii. $\cos\frac{2\pi}{9}\cos\frac{4\pi}{9}\cos\frac{\pi}{9}$
    iii. $\sec\frac{2\pi}{9} + \sec\frac{4\pi}{9} - \sec\frac{\pi}{9}$
    iv. $\cos^2\frac{2\pi}{9} + \cos^2\frac{4\pi}{9} + \cos^2\frac{\pi}{9}$

17. If α, β and γ are the roots of the equation $x^3 + 5x - 4 = 0$, evaluate $\alpha^3 + \beta^3 + \gamma^3$.

18. Suppose that the polynomial $P(x) = ax^3 + bx^2 + cx + d$ has zeroes α, β and γ.
    a. Show that $\alpha^2 + \beta^2 + \gamma^2 = \frac{b^2 - 2ac}{a^2}$
    b. Hence explain why the polynomial $2x^3 - 3x^2 + 5x - 8$ cannot have three real zeroes.

19. Suppose that the equation $x^3 + ax^2 + bx + c = 0$ has roots α, β and γ. If $\gamma = \alpha + \beta$, show that $a^3 - 4ab + 8c = 0$.

Answer

1. a. 4; b. 2; c. 8; d. 2; e. 14; f. 12; g. 6; h. 24; i. $\frac{17}{2}$
2. a. -2; b. -11; c. 12; d. $-\frac{11}{12}$; e. $-\frac{1}{6}$; f. 0; g. -132; h. 26; i. $\frac{13}{72}$; The roots are $-1, -4$ and 3
3. a. 5; b. 2; c. 4; d. -3; e. $-\frac{4}{3}$; f. $-\frac{2}{3}$; g. $-\frac{5}{3}$; h. 21
4. a. $-\frac{5}{2}$; b. -2; c. $\frac{41}{4}$; d. $-30\frac{5}{8}$; e. $\frac{1}{2}\sqrt{57}$
5. c. $\alpha^2 - 3\alpha + 5 = 0$, which has negative discriminant; d. once
6. a. $a = 5, b = 12, (x-3)(x+1)(x-4)$; b. $a = -5$, $b = 8, \frac{1}{2}$ is the other zero
7. a. $a = 3$ and $b = -24, (x-3)(x+4)(x+2)$; b. $a = -1$ and $b = 3, 5, -4, \sqrt{3}, -\sqrt{3}$
8. prove
9. prove
10. a. $\frac{3}{2}, \frac{3}{2}$ and -1; b. $\frac{1}{3}, -4$ and 4; c. $6, \frac{1}{2}$ and -4; d. $4, \frac{1}{2}$ and 2
11. a. $-\frac{1}{3}$, 1 and $\frac{7}{3}$. The inflexion is $(1, 0)$; b. $\frac{1}{4}, \frac{1}{2}$ and 1; c. $x = \frac{1}{2}, -1$ or 2
12. a. $a = -12$ and the roots are $-2, 2$ and -3; b. $a = -5$ and the roots are $4, \frac{1}{4}$ and -3; c. $-1, -2, 2$ and 4.
13. prove
14. prove
15. a. i. $\frac{1}{2}$, ii. $\frac{1}{2}$, iii. $\frac{1}{2}$, iv. $\frac{1}{4}$
16. c. $\cos\frac{2\pi}{9}, \cos\frac{4\pi}{9}, \cos\frac{8\pi}{9}$; d. i. 0, ii. $\frac{1}{8}$, iii. 6, iv. $\frac{3}{2}$
17. 12
18. b. $\alpha^2 + \beta^2 + \gamma^2 = -\frac{11}{4} < 0$ which is impossible if α, β and γ are all real numbers, because squares are never negative.
19. prove